

МИНИСТЕРСТВО НАУКИ И ВЫСШЕГО ОБРАЗОВАНИЯ
РОССИЙСКОЙ ФЕДЕРАЦИИ
Федеральное государственное автономное образовательное учреждение
высшего образования
УРАЛЬСКИЙ ФЕДЕРАЛЬНЫЙ УНИВЕРСИТЕТ
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Лабораторная работа №6
Тема: «Введение в многопоточность»

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Цель работы

Получить представление о механизме многопоточности в языке программирования Java

Ход выполнения

1

```
1 package lr6;
2
3 import java.time.LocalDateTime;
4
5 public class Task_1 {
6     public static void main(String[] args) throws InterruptedException {
7         Thread t1 = new Thread() -> {
8             for (int i = 0; i < 10; i++) {
9                 System.out.println(Thread.currentThread().getName() + " i=" + i);
10                try {
11                    Thread.sleep(1000);
12                } catch (InterruptedException e) {
13                    e.printStackTrace();
14                }
15            }
16        };
17
18        Thread t2 = new Thread() -> {
19            for (int i = 0; i < 10; i++) {
20                System.out.println(Thread.currentThread().getName() + " i=" + i);
21                try {
22                    Thread.sleep(1000);
23                } catch (InterruptedException e) {
24                    e.printStackTrace();
25                }
26            }
27        };
28
29        t1.start();
30        t2.start();
31
32        t1.join();
33        t2.join();
34    }
35 }
```

/Users/narseev/Library/Java/JavaVirtualMachine-11.0.2/...
Thread-1:16:59:43.221036
Thread-0:16:59:43.221040
Thread-0:16:59:44.226686
Thread-1:16:59:44.226712
Thread-1:16:59:45.228659
Thread-0:16:59:45.230116
Thread-1:16:59:46.232901
Thread-0:16:59:46.233007
Thread-0:16:59:47.238133
Thread-1:16:59:47.238133
Thread-1:16:59:48.242474
Thread-0:16:59:48.243558
Thread-0:16:59:49.246841
Thread-1:16:59:49.246841
Thread-0:16:59:50.251939
Thread-1:16:59:50.251939
Thread-0:16:59:51.253183
Thread-1:16:59:51.254521
Thread-0:16:59:52.256634
Thread-1:16:59:52.256634
Process finished with exit code 0

2

```
1 package lr6;
2
3 public class Task_2 {
4     public static void main(String[] args) throws InterruptedException {
5         Thread t = new Thread() -> {
6             for (int i = 1; i <= 10; i++) {
7                 System.out.println(i);
8                 try {
9                     Thread.sleep(1000);
10                } catch (InterruptedException e) {
11                    e.printStackTrace();
12                }
13            }
14        };
15
16        t.start();
17        t.join();
18    }
19 }
```

/Users/narseev/Library/Java/JavaVirtualMachine-11.0.2/...
1
2
3
4
5
6
7
8
9
10
Process finished with exit code 0

3

```

1 package lr6;
2
3 public class Task_3 {
4     private static final Object lock = new Object(); 6 usages
5     private static int number = 1; 8 usages
6
7     public static void main(String[] args) {
8         Thread evenThread = new Thread(() -> {
9             synchronized (lock) {
10                 while (number < 10) {
11                     if (number % 2 == 0) {
12                         System.out.println(Thread.currentThread().getName() + "Even Thread:" + number);
13                         number++;
14                     } else {
15                         try {
16                             lock.wait();
17                         } catch (InterruptedException e) {
18                             e.printStackTrace();
19                         }
20                     }
21                 }
22             }
23         });
24         Thread oddThread = new Thread(() -> {
25             while (number < 10) {
26                 System.out.println(Thread.currentThread().getName() + "Odd Thread:" + number);
27                 number++;
28             }
29         });
30         evenThread.start();
31         oddThread.start();
32     }
33 }

```

Process finished with exit code 0

4

```

1 package lr6;
2
3 public class Task_4 {
4     public static void main(String[] args) {
5         Thread[] threads = new Thread[10];
6
7         for (int i = 0; i < threads.length; i++) {
8             int threadNumber = i + 1;
9             threads[i] = new Thread(() -> System.out.println("Номер потока: " + threadNumber));
10            threads[i].start();
11        }
12
13        for (Thread thread : threads) {
14            try {
15                thread.join();
16            } catch (InterruptedException e) {
17                Thread.currentThread().interrupt();
18                break;
19            }
20        }
21    }
22 }

```

Process finished with exit code 0

5

```

/Users/narseev/Library/Java/JavaVirtualMachines/openjdk-
Массив: [12, 4, 56, 7, 89, 34, 3, 150, 2, 77, 64, 101]
Максимум: 150

Process finished with exit code 0

```

```
/Users/narseev/Library/Java/JavaVirtualMachines/openjdk-  
Массив: [12, 4, 56, 7, 89, 34, 3, 150, 2, 77, 64, 101]  
Сумма: 599  
  
Process finished with exit code 0
```

GitHub

Ссылка: https://github.com/gh-u14/java_labs